

## WATERLOO, IA, USA

WMO: 725480

Lat: 42.554N

Lon: 92.401W

Elev: 265

StdP: 98.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94910

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -23.3      | -20.4 | -26.9                         | 0.3 | -22.7 | -24.2 | 0.4 | -19.9 | 12.9                  | -7.1 | 11.8 | -7.9 | 3.1                   | 300  | 0.588 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 32.7            | 24.2 | 31.2 | 23.2 | 29.7 | 22.4 | 25.9                | 30.5 | 24.8 | 29.2 | 23.8 | 28.1 | 5.6                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.5                            | 20.2 | 29.2 | 23.4 | 18.7 | 27.7 | 22.3 | 17.5 | 26.6 | 81.6          | 30.4 | 76.4 | 29.2 | 72.3 | 28.1 | 29.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 11.6              | 10.4 | 9.0 | -28.6                      | 35.3 | 4.2                | 1.7 | -31.6                                              | 36.5 | -34.0      | 37.5 | -36.4      | 38.4 | -39.4      | 39.6 |
|                   |      |     | -28.8                      | 27.5 | 4.1                | 1.2 | -31.7                                              | 28.3 | -34.1      | 29.0 | -36.4      | 29.6 | -39.4      | 30.5 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 8.9    | -7.6  | -5.2  | 1.9   | 9.1   | 15.7  | 21.2  | 22.9  | 21.5  | 17.7  | 10.4  | 2.6   | -4.3 |
|                                                                               | DBStd       | 12.00  | 7.23  | 7.13  | 6.75  | 5.43  | 4.68  | 3.48  | 3.17  | 3.02  | 4.54  | 5.30  | 5.88  | 6.70 |
|                                                                               | HDD10.0     | 2060   | 547   | 427   | 264   | 78    | 7     | 0     | 0     | 0     | 2     | 60    | 231   | 444  |
|                                                                               | HDD18.3     | 3876   | 805   | 659   | 511   | 279   | 110   | 13    | 3     | 6     | 65    | 252   | 472   | 702  |
|                                                                               | CDD10.0     | 1659   | 0     | 1     | 13    | 52    | 183   | 337   | 400   | 358   | 234   | 73    | 9     | 1    |
|                                                                               | CDD18.3     | 434    | 0     | 0     | 1     | 3     | 27    | 99    | 144   | 106   | 47    | 6     | 0     | 0    |
|                                                                               | CDH23.3     | 4144   | 0     | 0     | 7     | 56    | 288   | 927   | 1398  | 925   | 478   | 64    | 1     | 0    |
|                                                                               | CDH26.7     | 1336   | 0     | 0     | 0     | 13    | 80    | 312   | 496   | 272   | 149   | 14    | 0     | 0    |
| Wind                                                                          | WSAvg       | 4.2    | 4.6   | 4.7   | 4.9   | 5.3   | 4.6   | 3.9   | 3.3   | 3.0   | 3.6   | 4.2   | 4.5   | 4.4  |
| Precipitation                                                                 | PrecAvg     | 874    | 19    | 27    | 58    | 89    | 116   | 113   | 125   | 100   | 82    | 65    | 49    | 30   |
|                                                                               | PrecMax     | 1348   | 45    | 90    | 138   | 217   | 288   | 257   | 326   | 268   | 289   | 138   | 142   | 96   |
|                                                                               | PrecMin     | 482    | 2     | 0     | 6     | 24    | 21    | 36    | 8     | 2     | 14    | 1     | 2     | 3    |
|                                                                               | PrecStd     | 191    | 12    | 20    | 31    | 42    | 60    | 51    | 79    | 63    | 60    | 36    | 36    | 20   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 10.7  | 13.9  | 23.9  | 28.7  | 31.9  | 33.8  | 35.5  | 33.6  | 32.6  | 28.7  | 21.3  | 14.3 |
|                                                                               |             | MCWB   | 6.1   | 9.7   | 15.9  | 17.8  | 21.9  | 24.3  | 25.5  | 24.0  | 22.8  | 19.8  | 13.8  | 11.3 |
|                                                                               | 2%          | DB     | 6.0   | 9.8   | 19.0  | 24.6  | 28.8  | 31.7  | 32.9  | 31.4  | 30.3  | 24.8  | 17.4  | 9.7  |
|                                                                               |             | MCWB   | 3.4   | 6.8   | 13.1  | 15.6  | 19.3  | 22.8  | 25.3  | 24.1  | 21.7  | 17.1  | 12.6  | 6.7  |
|                                                                               | 5%          | DB     | 3.4   | 6.4   | 15.5  | 21.5  | 26.6  | 30.2  | 31.1  | 29.5  | 28.3  | 22.0  | 14.2  | 6.9  |
|                                                                               |             | MCWB   | 1.8   | 4.1   | 11.1  | 13.5  | 18.1  | 21.9  | 24.3  | 23.0  | 20.5  | 16.3  | 10.1  | 4.8  |
|                                                                               | 10%         | DB     | 1.5   | 3.8   | 11.7  | 18.8  | 24.3  | 28.3  | 29.5  | 28.0  | 26.2  | 19.4  | 11.5  | 4.1  |
|                                                                               |             | MCWB   | 0.3   | 2.1   | 8.0   | 12.1  | 16.9  | 20.8  | 23.3  | 22.1  | 19.2  | 14.5  | 7.9   | 2.4  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 6.5   | 10.3  | 17.0  | 18.8  | 23.2  | 26.4  | 27.8  | 26.7  | 24.5  | 21.3  | 15.7  | 12.3 |
|                                                                               |             | MCDB   | 10.2  | 13.4  | 22.1  | 25.9  | 29.4  | 32.1  | 32.6  | 31.1  | 30.3  | 25.7  | 19.5  | 13.8 |
|                                                                               | 2%          | WB     | 3.8   | 7.0   | 14.6  | 16.7  | 21.1  | 24.6  | 26.4  | 25.2  | 23.0  | 19.1  | 13.3  | 7.7  |
|                                                                               |             | MCDB   | 5.9   | 9.4   | 18.0  | 22.5  | 26.7  | 29.4  | 31.1  | 29.6  | 28.1  | 22.7  | 16.4  | 9.1  |
|                                                                               | 5%          | WB     | 1.8   | 4.1   | 11.1  | 14.9  | 19.7  | 23.3  | 25.3  | 24.1  | 21.8  | 17.1  | 10.6  | 4.6  |
|                                                                               |             | MCDB   | 3.2   | 6.1   | 15.0  | 20.4  | 24.6  | 28.3  | 29.6  | 27.9  | 26.1  | 20.8  | 13.8  | 6.7  |
|                                                                               | 10%         | WB     | 0.5   | 2.1   | 8.0   | 12.7  | 18.2  | 22.1  | 24.1  | 23.0  | 20.5  | 15.1  | 8.2   | 2.5  |
|                                                                               |             | MCDB   | 1.5   | 3.6   | 11.4  | 17.6  | 22.7  | 27.0  | 28.3  | 26.6  | 24.6  | 18.8  | 11.1  | 4.0  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.1   | 9.2   | 10.2  | 12.1  | 11.7  | 11.1  | 11.1  | 11.3  | 12.9  | 11.8  | 10.2  | 8.7  |
|                                                                               |             | MCDBR  | 11.3  | 11.3  | 14.9  | 16.8  | 14.5  | 13.0  | 12.0  | 12.3  | 14.6  | 15.0  | 14.5  | 11.6 |
|                                                                               |             | MCWBR  | 9.1   | 8.6   | 9.3   | 9.4   | 7.0   | 6.0   | 5.9   | 6.2   | 6.9   | 8.4   | 9.7   | 9.1  |
|                                                                               | 5% WB       | MCDBR  | 10.4  | 10.6  | 13.8  | 14.3  | 12.4  | 11.3  | 10.9  | 10.6  | 11.8  | 12.4  | 12.9  | 10.7 |
|                                                                               |             | MCWBR  | 8.5   | 8.3   | 9.0   | 8.9   | 6.8   | 6.2   | 6.0   | 6.1   | 6.4   | 8.3   | 9.4   | 8.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.284  | 0.309 | 0.332 | 0.376 | 0.406 | 0.418 | 0.434 | 0.415 | 0.384 | 0.340 | 0.301 | 0.279 |      |
|                                                                               | taud        | 2.379  | 2.313 | 2.372 | 2.281 | 2.266 | 2.276 | 2.215 | 2.281 | 2.328 | 2.446 | 2.481 | 2.421 |      |
|                                                                               | Ebn at Noon | 860    | 890   | 912   | 890   | 869   | 856   | 837   | 843   | 844   | 846   | 837   | 835   |      |
|                                                                               | Edh at Noon | 79     | 100   | 107   | 127   | 133   | 132   | 139   | 126   | 112   | 88    | 72    | 70    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.85   | 2.89  | 3.85  | 4.67  | 5.55  | 6.13  | 6.23  | 5.40  | 4.45  | 2.91  | 1.97  | 1.46  |      |
|                                                                               | RadStd      | 0.18   | 0.24  | 0.40  | 0.41  | 0.39  | 0.53  | 0.36  | 0.35  | 0.32  | 0.37  | 0.16  | 0.11  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (53 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page